

PART - A

Q1. Bagging.

* Bagging is an ensemble learning technique which combines multiple dataset using models. The models are trained using different subsets of data parallelly.

Example:

* Random Forest uses multiple decision trees.

Q2. Random Forest reduces variance.

* Random Forest uses multiple decision trees and predicts the output based on voting methods leads to reduce of variance.

Q3. k-Means clustering.

* k-means clustering is unsupervised algorithm which groups the data points based on the centroids.

* The process is repeated until the no change in centroid.

Q4. Centroid updation:

* Centroid is updated by calculating the mean value for all datapoints assigned for the specific cluster.

Q5.

PCA : (Principal component Analysis)

* Principal component analysis is one of the dimensionality reduction technique.

* PCA converts the correlated data variables into smaller number of correlated variables.

Q6.

Heirarchical

Clustering

* Heirarchical clustering is represented using dendrogram

* Initially, each data point is considered as own centroid.

* Used for smaller datasets

k-means.

* It is not represented using dendrogram.

* k' value is selected, centroids are chosen randomly.

* Used for larger datasets.